A Project Synopsis

On

SehatBuddy

For the Degree of
Bachelor of Technology

In

Computer Science and Engineering

By

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1. Project Domain

This project falls under the broader domain of **Artificial Intelligence in Healthcare**, with a specialized focus on wearable-based health monitoring and anomaly detection. Specifically focusing on anomaly detection in wearable-generated time-series data. The healthcare industry is rapidly shifting from hospital-centered care to continuous, patient-centric monitoring through smart devices and mobile technologies. Wearable devices such as smartwatches and fitness bands now generate large volumes of physiological and motion-related data in real time.

Unlike traditional clinical systems that rely on fixed medical thresholds, this project explores intelligent anomaly detection models that learn a user's normal behavioral and physiological patterns and detect deviations from that personalized baseline.

The wearable domain specifically includes:

- Motion signals (accelerometer, gyroscope)
- Heart rate trends
- Activity levels and step patterns
- Sleep duration and quality
- Context-aware physiological variations

The anomalies associated within this domain include:

- Sudden fall events
- Abnormal movement spikes
- Extended inactivity
- Elevated resting heart rate
- Irregular sleep reduction
- Stress-related physiological deviations
- Behavioral pattern shifts over time

The core idea of this domain is not just monitoring raw data, but converting continuous wearable sensor streams into meaningful health insights using AI models. The system focuses on humanized health intelligence — meaning the alerts and decisions are personalized to each individual rather than based on rigid medical rules.

By integrating smartphone sensors, wearable devices, and machine learning inference, the project contributes toward proactive healthcare monitoring, early anomaly detection, and real-time health awareness.

2. Literature Survey / Related Work/ Company provided reference material

Machine Learning Algorithms for Wearable Anomaly Detection

Anomaly detection in wearable-generated healthcare time-series data has been widely studied using both classical machine learning and deep learning approaches. These methods aim to identify abnormal patterns in multivariate sensor streams such as accelerometer, gyroscope, and physiological signals.

Classical unsupervised models such as One-Class Support Vector Machines (OC-SVM) and Local Outlier Factor (LOF) have been commonly used for detecting statistical deviations in sensor data due to their simplicity and low computational cost. However, their performance is limited in handling long-term temporal dependencies in sequential data [2]. Similarly, Isolation Forest has been widely applied for anomaly detection in healthcare due to its efficiency and ability to handle high-dimensional data [1].

To better capture temporal dependencies in time-series data, deep learning methods have gained significant attention. Autoencoders and Variational Autoencoders (VAEs) detect anomalies using reconstruction error and have been widely used in healthcare anomaly detection systems [1].
Sequential deep learning architectures such as CNN-LSTM combine convolutional feature extraction with temporal modeling and have shown improved performance in human activity recognition and fall detection tasks [3].
Temporal Convolutional Networks (TCNs) offer advantages over recurrent models by capturing long-range dependencies with lower computational overhead, making them suitable for mobile deployment [1].
Recent studies have also explored Transformer-based models for capturing global dependencies in long time-series data [1].

Wearable datasets such as MobiFall [4], USC-HAD [5], and StressSense [6] have been extensively used to evaluate these models. These datasets provide multivariate sensor streams that enable the development of robust anomaly detection frameworks for real-world healthcare applications.

Overall, literature suggests that deep learning models outperform classical techniques in multivariate wearable anomaly detection, especially when temporal dependencies and multi-sensor fusion are considered [1].

Fitness Bands and Wearable Devices

Wearable fitness bands and smartwatches play a crucial role in continuous health monitoring by collecting physiological and motion data in real time. Modern fitness bands are equipped with multiple sensors such as accelerometers, gyroscopes, optical heart rate

monitors, and sleep tracking modules.

1. Mi Smart Band 8 (Xiaomi)

- Features: Heart rate monitoring, SpO_2 tracking, sleep tracking, activity detection, 150+ sports modes
- Sensors: Accelerometer, PPG heart rate sensor
- Battery Life: Up to 14 days
- Price: ₹3,000 – ₹4,000 (India)

2. Fitbit Charge 6

- Features: Continuous heart rate tracking, ECG support, stress tracking, sleep score, Google integration.
- Sensors: Heart rate, ECG, SpO_2, motion sensors
- Battery Life: ~7 days
- Price: ₹14,000 – ₹18,000

3. Samsung Galaxy Fit 3

- Features: Activity tracking, sleep monitoring, fall detection, stress tracking
- Sensors: Accelerometer, gyroscope, heart rate sensor
- Battery Life: Up to 13 days
- Price: ₹5,000 – ₹7,000

4. Apple Watch SE (2nd Gen)

- Features: Heart rate monitoring, fall detection, crash detection, activity tracking, sleep tracking
- Sensors: Accelerometer, gyroscope, optical heart rate sensor
- Battery Life: ~18 hours
- Price: ₹25,000 – ₹30,000

These devices generate large volumes of real-time physiological and motion data, making them suitable for AI-driven health monitoring and anomaly detection systems.

Mobile Applications for Fitness and Health Tracking

Several mobile applications are available that integrate wearable devices and provide fitness tracking and health insights. However, most existing apps rely on rule-based alerts rather than intelligent anomaly detection.

1. Google Fit

- Tracks: Steps, heart rate, calories, activity duration
- Integration: Android wearables and third-party fitness devices
- Features: Health metrics dashboard, activity goals, wearable integration

2. Apple Health

- Tracks: Heart rate, sleep, mobility metrics, workouts
- Integration: Apple Watch ecosystem
- Features: Health records integration, activity rings, medical alerts

3. Fitbit App

- Tracks: Heart rate, sleep stages, stress levels, readiness score
- Features: Personalized insights, guided workouts, wellness reports
- Strong integration with Fitbit devices

4. Samsung Health

- Tracks: Activity, sleep, heart rate, stress
- Features: Health dashboard, fitness programs, wearable syncing
- Popular among Samsung wearable users

5. Strava

- Focus: Activity tracking for runners and cyclists
- Features: GPS tracking, performance analytics, community challenges
- Supports smartwatch integrations

While these apps provide extensive health monitoring features, most rely on fixed thresholds and basic analytics. There is still a gap in **personalized anomaly detection**, where the system learns individual **behavioral baselines** and detects deviations using AI models.

3. Problem Statement

Wearable devices and smartphones continuously generate large volumes of time-series health and motion data such as accelerometer readings, gyroscope signals, heart rate trends, sleep cycles, and activity patterns. Although this data has strong potential for early health monitoring and anomaly detection, most existing applications rely primarily on simple activity tracking or fixed threshold-based alerts. These approaches do not intelligently learn a user's normal behavioral patterns and fail to effectively analyze multivariate time-series data in real time.

Existing research in wearable anomaly detection has explored various machine learning and deep learning models for identifying abnormal physiological and motion patterns. However, most of these models are designed and evaluated under static assumptions, where the definition of "normal behavior" remains constant over time. In real-world scenarios, human behavior naturally evolves due to lifestyle changes, aging, recovery from illness, seasonal variations, and fitness improvements. This phenomenon is known as concept shift (or concept drift), where the statistical properties of data change over time.

Most existing anomaly detection systems do not explicitly detect or adapt to concept shift. As a result, they may generate increased false alarms or miss important anomalies when the user's baseline behavior changes. While previous research phases have demonstrated strong anomaly detection performance on benchmark wearable datasets, there remains a significant gap in translating these models into adaptive real-world systems capable of handling evolving behavioral patterns.

Therefore, the problem addressed in this project is the design and implementation of a real-time, personalized anomaly detection system with concept shift awareness that can collect wearable sensor data, analyze multivariate time-series streams on-device, detect both anomalies and behavioral drift, and generate meaningful, context-aware alerts while maintaining computational efficiency, privacy, and low power consumption.

4. Objectives of Study

- To collect and integrate data from handheld and wearable devices using research-based models for accurate and reliable health anomaly detection.
- To design a smart system that safely connects phone and watch sensors to collect health data privately.
- To develop a clever app that learns your body's normal patterns and ignores minor changes to avoid false alarms.
- To test the app thoroughly to make sure it catches real health issues without slowing down the phone or draining the battery.
- To deploy a finished tool that gives instant health alerts while keeping all your personal information locked on your device.

5. Proposed Approach/Methodology

The project will follow a practical approach where the mobile app and the anomaly detection model are developed side by side. This helps in making sure that the model works properly with real-time sensor data collected from the phone and wearable devices.

Step 1: Understanding Previous Research

First, we study the datasets we used in the previous semester such as PAMAP2 Physical Activity Monitoring Dataset, MobiFall Dataset, USC-HAD Dataset and StressSense Dataset.
We understand what type of sensor data they used (accelerometer, gyroscope, heart rate, etc.) and how anomalies were detected.

Step 2: Collecting Real-Time Data

The app will collect motion data like accelerometer and gyroscope from the smartphone. It will also collect health data like heart rate, steps, and sleep from wearable devices using proper permissions. We make sure the format of this live data matches what our model expects.

Step 3: Preparing the Data

The raw sensor data may have noise, so we clean it.
Then we divide the continuous data into small time windows (for example, 3–5 seconds). This makes it easier for the model to analyze patterns.

Step 4: Using the Best Model from Previous Research

From our last semester research, we already tested many machine learning and deep learning models.
Instead of using all models again, we select the best-performing and most efficient one. We optimize it so it can run smoothly on a mobile phone.

Step 5: Integrating Model into the App

The selected model is deployed on a secure server rather than directly inside the mobile application.The app collects live sensor data, preprocesses it, and securely sends it to the server through an API.The server runs the model and returns an anomaly score in real time.Only the results are sent back to the mobile app, ensuring efficiency, scalability, and data privacy.

Step 6: Generating Alerts

If the anomaly score crosses a certain level, the app generates alerts. Alerts can be like possible fall detection, unusual inactivity, abnormal heart rate, or sleep reduction. The app will give options like dismiss alert or contact emergency.

Step 7: Testing and Improvement

We test the system properly to reduce false alarms.
We adjust thresholds and improve efficiency so the app does not consume too much battery. Finally, we make sure the system works smoothly in real-world conditions.

Flow chart:

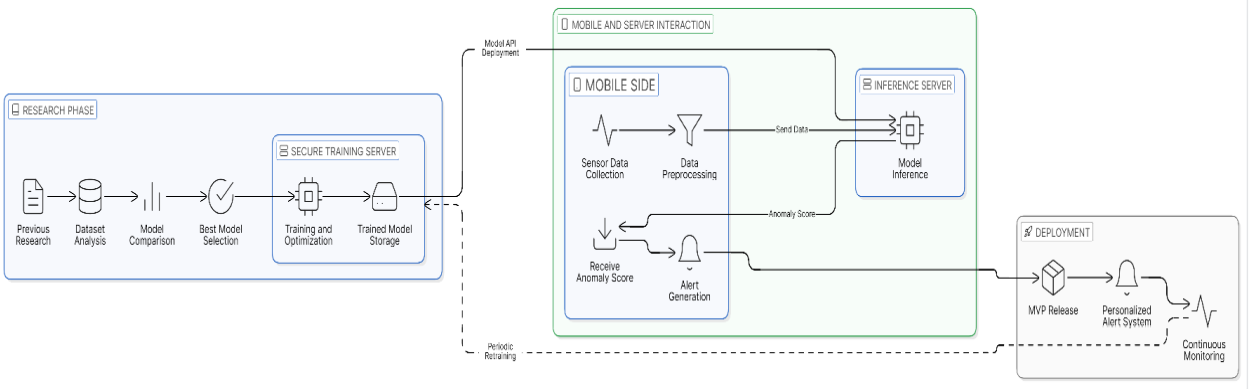

Fig 1.1 Flowchart of proposed methodology

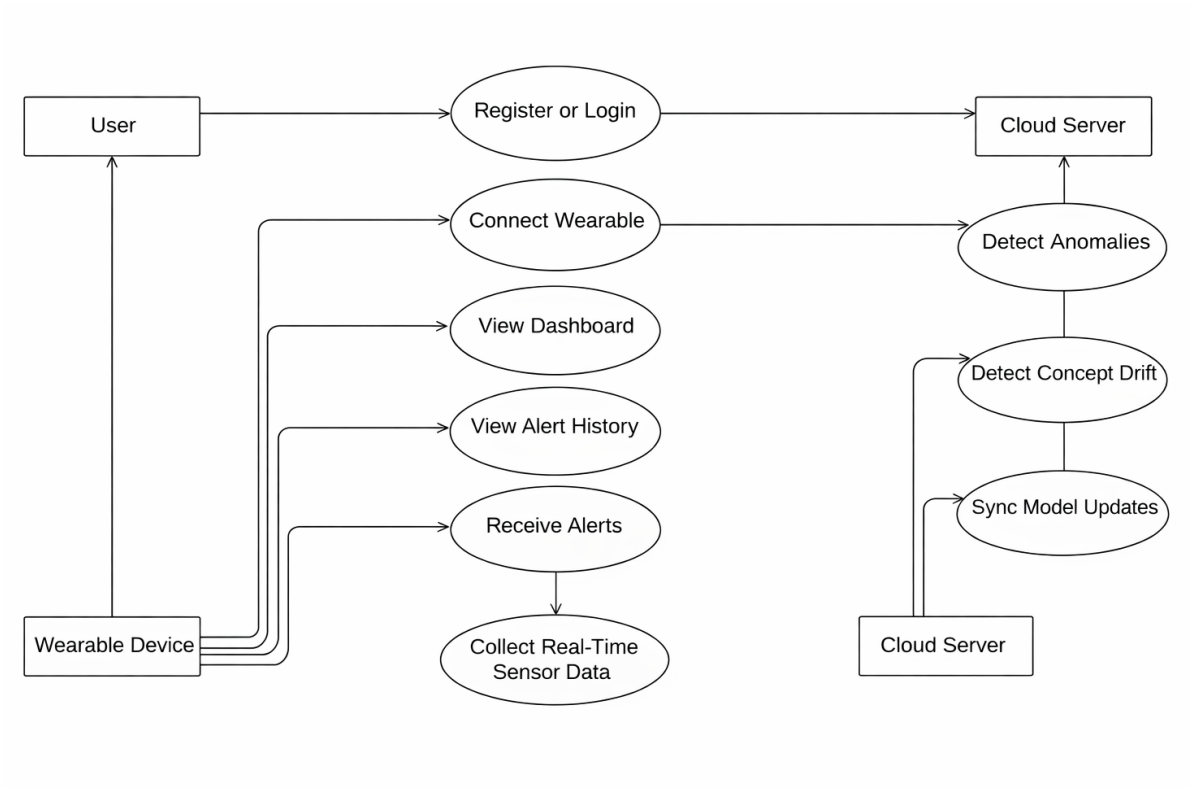

Fig 1.2 Use Case Diagram

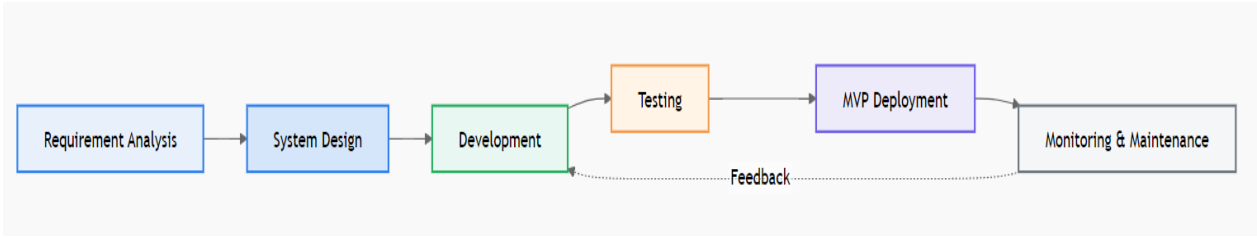

Fig 1.3 App development flowchart

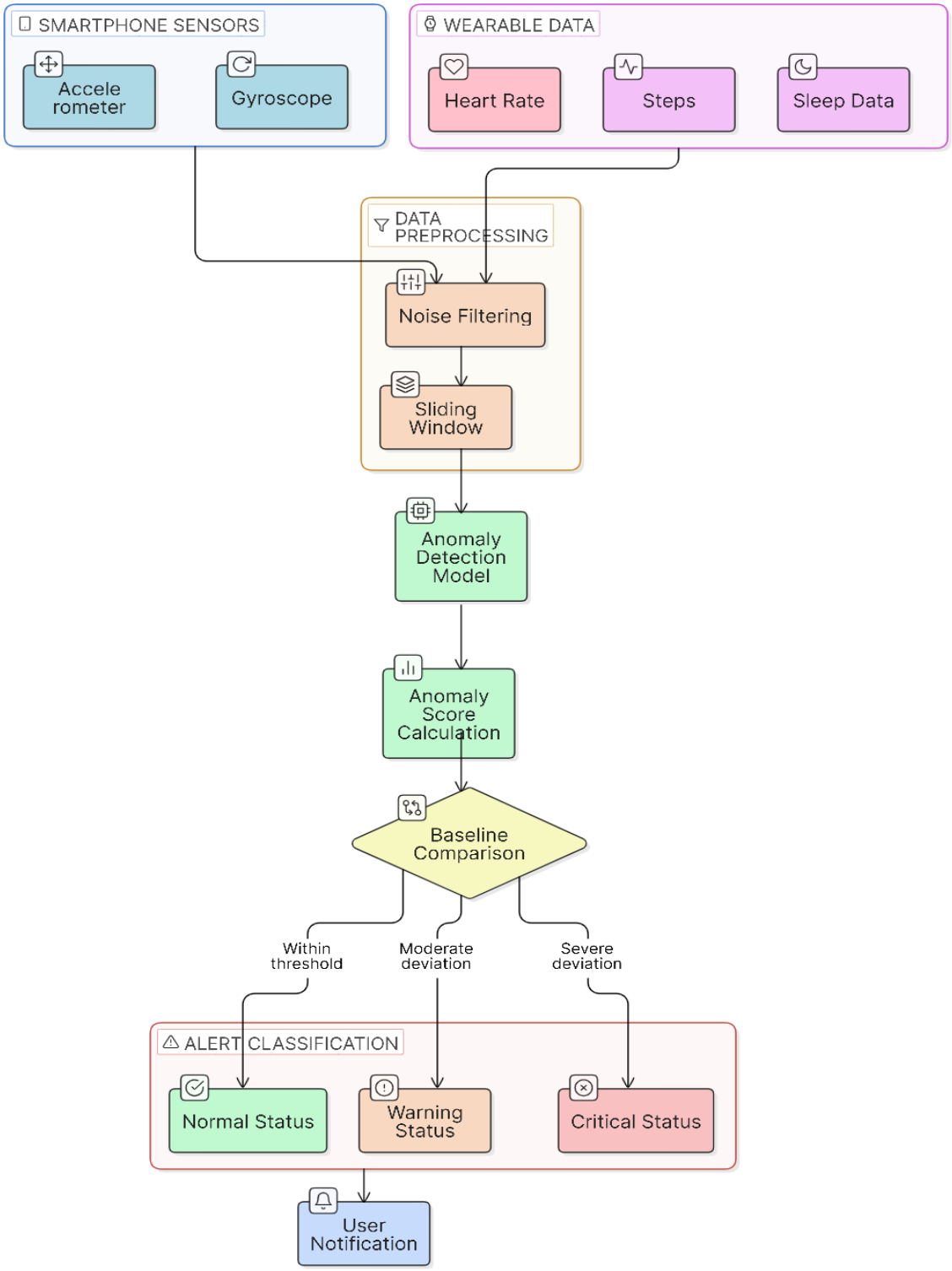

Fig 1.4 Flowchart of app

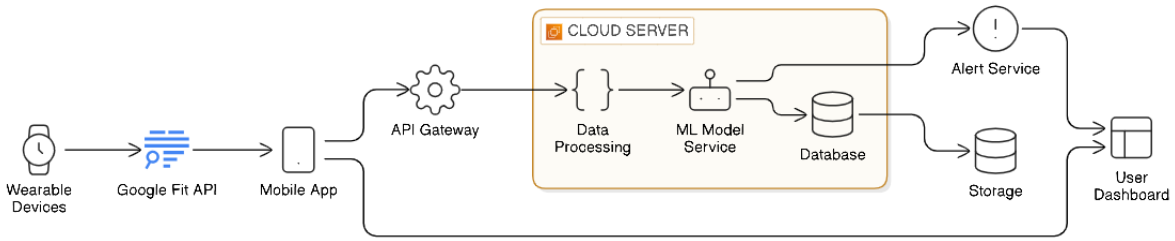

Fig 1.5 Architecture of application

6. Proposed/ Expected outcomes/ Deliverables

The proposed project is expected to deliver a working mobile application that collects real-time sensor data from smartphones and wearable devices and intelligently analyzes it using an anomaly detection model trained from previous research. The system will detect abnormal patterns in user activity, movement, or heart rate and generate timely alerts based on deviations from personal baseline behavior.

The application can be practically useful in several ways:

- Early detection of unusual health conditions such as sudden inactivity, abnormal motion, or irregular heart rate patterns.
- Support for fall detection and abnormal activity recognition in daily life.
- Continuous health monitoring without requiring hospital supervision.
- Personalized health tracking based on individual baseline patterns.
- Providing users with real-time alerts and health status insights through a simple dashboard.

Overall, the project is expected to demonstrate how wearable-based AI systems can serve as a preventive healthcare support tool, making health monitoring more accessible, intelligent, and proactive.

The Novelty of the Project Will Be:

1. Personalized Anomaly Detection – The system learns each user's normal behavior pattern and detects deviations from their personal baseline instead of using fixed medical thresholds.
2. Multi-Modal Sensor Fusion – Motion data and physiological data are analyzed together to improve anomaly detection accuracy.
3. Research-to-Real-World Integration – The project converts research-validated anomaly detection models into a real-time deployable mobile application prototype
4. Adaptive Alert Generation – Alerts are triggered using dynamic anomaly scores rather than hard-coded health rules.

Deliverables

1. Dataset Analysis and Model Evaluation Report
   A detailed report presenting preprocessing, feature analysis, and performance comparison of anomaly detection models on wearable time-series data.

2. Optimized Anomaly Detection Model
   A finalized and trained machine learning/deep learning model selected from previous research and adapted for real-time wearable data.

3. Mobile Application Prototype
   A functional mobile application capable of collecting sensor data, running the anomaly detection model, and generating alerts based on abnormal patterns.

4. Project Documentation
   Complete documentation including methodology, system design, implementation details, model performance results, and application workflow.

7. Hardware/ Software specification

Hardware Requirements

- Smartphone (Android Device)
  For real-time data collection, model execution, and user interface.
- Wearable Device (Smart Band / Smartwatch)
  To collect physiological and motion data such as heart rate, steps, sleep, accelerometer, and gyroscope signals.
- Development System (Laptop/PC)
  For model training, testing, and application development.

Software Requirements

- Programming Language
  Python (for model development and server-side API implementation)
  Flutter (for mobile application development)
- Machine Learning Framework
  TensorFlow (for model training and server-side inference)
- Backend / Server Framework
  FastAPI / Flask (for building REST API for model inference)
  Uvicorn / Gunicorn (for API deployment)
- Development Tools
  VS Code
  Postman (for API testing)
- Data Processing Libraries
  NumPy, Pandas, Scikit-learn, Matplotlib
- Wearable Integration APIs

Google Fit / Health APIs (for wearable data access)

- Operating System
  Windows / Linux (for development and server deployment)
  Android OS (for mobile application deployment)

8. Project plan

Project Timeline (12 Weeks):

- Week 1–2: Literature review, concept drift study, and system architecture design.
- Week 3–4: Dataset analysis, real-time windowing, normalization, and feature extraction setup.
- Week 5–6: Lightweight anomaly detection model selection, training, evaluation, and mobile-friendly conversion.
- Week 7: Implementation of concept shift detection and adaptive anomaly scoring mechanism.
- Week 8–9: Mobile app development – UI/UX design, real-time sensor data collection, and preprocessing integration.
- Week 10: Integration of anomaly detection model and drift detection module into the mobile app with alert generation.
- Week 11: End-to-end testing, validation, and performance optimization (battery, latency, memory).
- Week 12: Final deployment build, documentation, and project review.

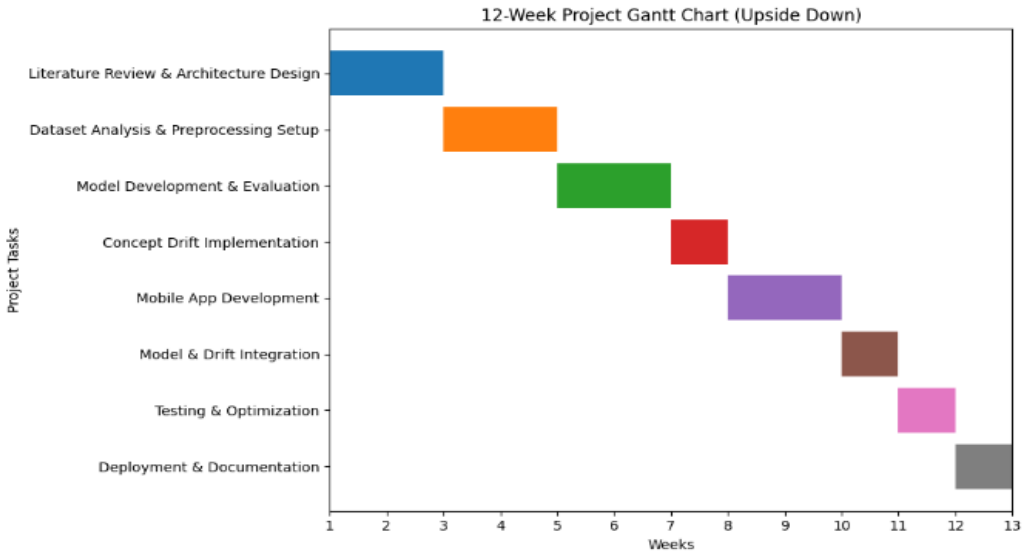

Fig 2.1 Gantt chart for project plan

9. References

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[7] Schmidt, P., Reiss, A., Duerichen, R., Marberger, C., & Van Laerhoven, K. (2018, October). Introducing wesad, a multimodal dataset for wearable stress and affect detection. In *Proceedings of the 20th ACM international conference on multimodal interaction* (pp. 400-408).

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